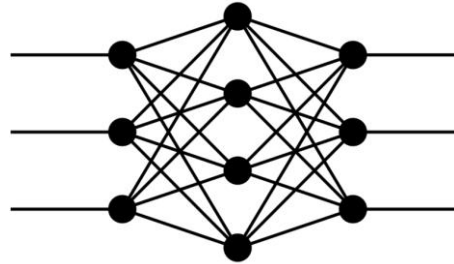
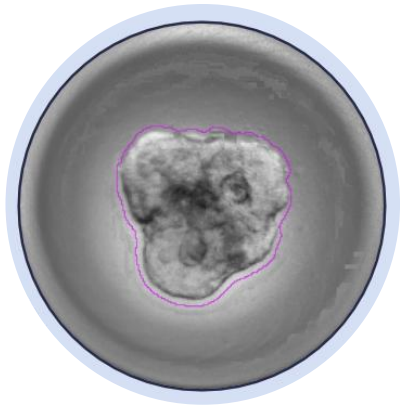
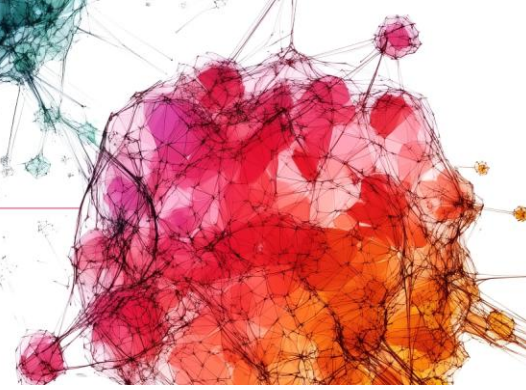


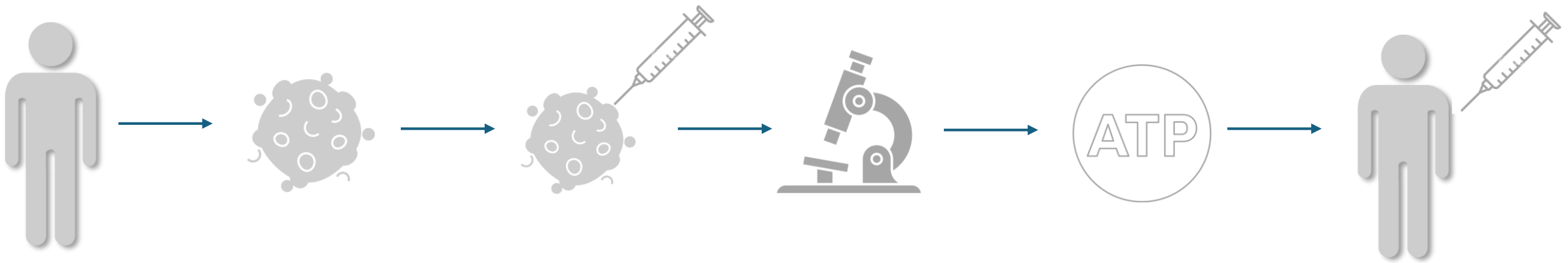


Deep Learning Project: ATP Prediction from Organoid Time-Lapse Data





Organoids and ATP: A Functional Approach to Personalized Cancer Treatment



Patient with
CGR or PGR

Patient derived
organoids

Application of
various treatments
to the organoids

Large-scale time-
lapse imaging of
organoids over 96
hours

ATP assay used to
assess drug
response

Personalized
drug

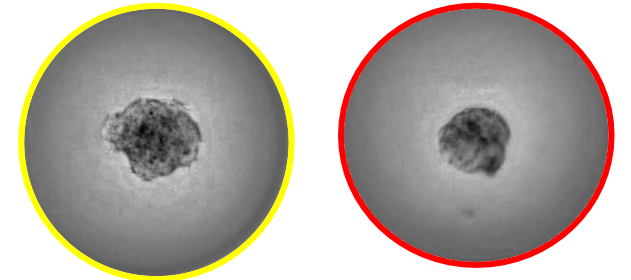


Introduction to Patients-Derived Organoids

« An organoid is a 3D miniaturized and simplified version of an organ grown in the lab from stem cells, which mimics key structural and functional features of the real organ. »

Organoids in Precision medicine :

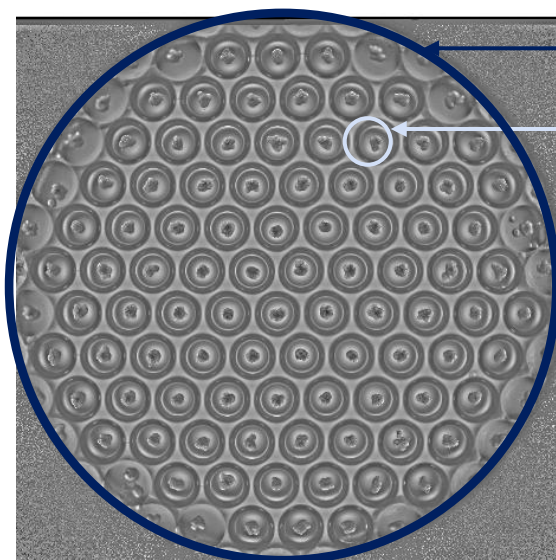
1. Predict drug response and resistance
2. Reduce trial-and-error in treatment selection
3. Support the development of personalized therapies tailored to each patient's biology





Dataset overview

- 92 632 cavity videos (8 frames) from 1 199 wells
- One well corresponds to a specific treatment at a given concentration for a particular patient
- ATP (a measure of cellular energy production) is only available at the well level.



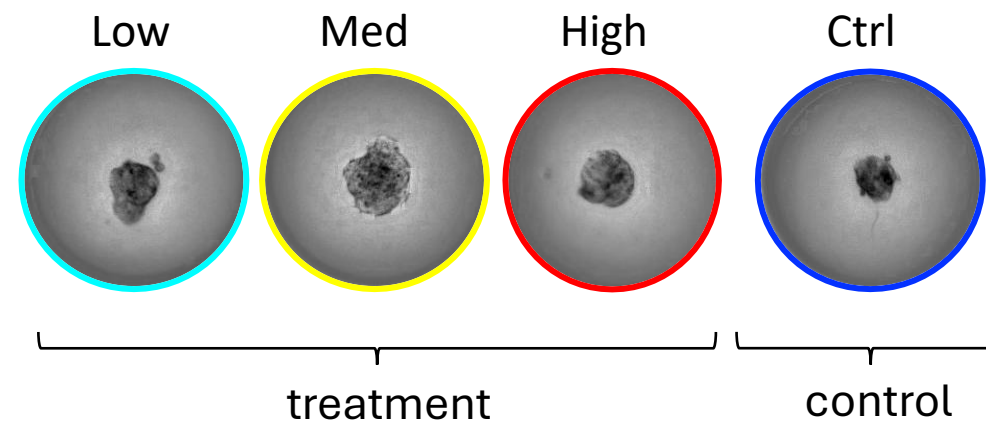
Well

Cavity

Patient cohort

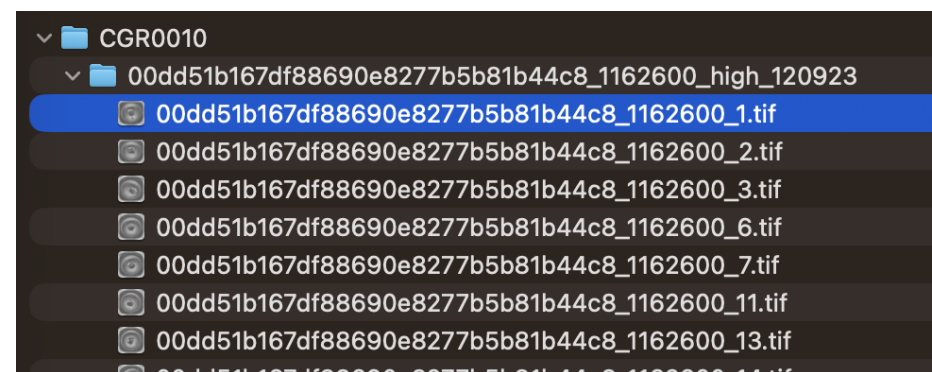
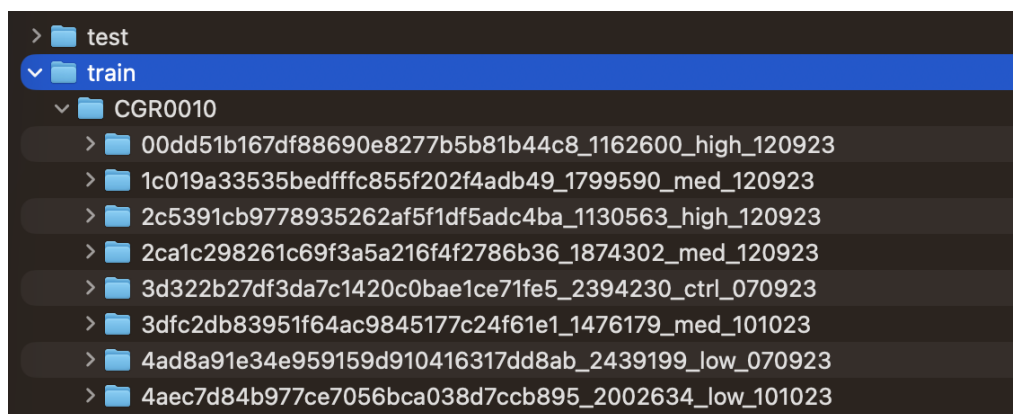
Number of patients	51
CGR	25
PGR	26

Patient treatment tested at 3 concentrations





Dataset structure



Train set : 36 patients

Test set : 15 patients

7a683f90345118ecd3dc6788de7dfea6_125952_high_40325



Unique well ID

concentration



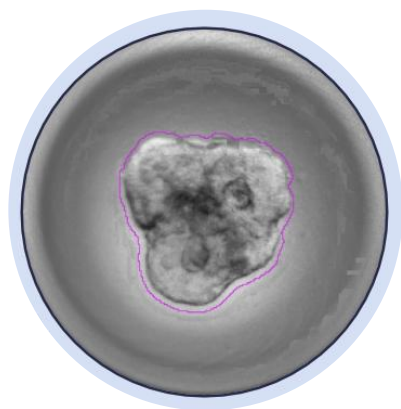
ATP



date



Pipeline for submission



Your model



Predictions submitted in the CSV file must be raw ATP values

sample_ID	ATP_prediction
0f5560338b74cc9ae8c6f6ae2c2f4003_med_180723	100000
7e55153bcd5380c039b99d5ec509f3b_low_250723	200000
0f5560338b74cc9ae8c6f6ae2c2f4003_med_180724	300000
7e55153bcd5380c039b99d5ec509f3b_low_250724	400000
0f5560338b74cc9ae8c6f6ae2c2f4003_med_180725	500000
7e55153bcd5380c039b99d5ec509f3b_low_250725	600000
0f5560338b74cc9ae8c6f6ae2c2f4003_med_180726	700000
7e55153bcd5380c039b99d5ec509f3b_low_250726	800000
0f5560338b74cc9ae8c6f6ae2c2f4003_med_180727	900000
7e55153bcd5380c039b99d5ec509f3b_low_250727	1000000
0f5560338b74cc9ae8c6f6ae2c2f4003_med_180728	1100000

submission file (.csv)



The ATP prediction will be evaluated using the Mean Absolute Percentage Error (MAPE).

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{actual_i - predicted_i}{actual_i} \right|$$

n = number of considered points

$actual_i$ = actual value

$predicted_i$ = the predicted value